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ENDOGENIC PROCESSES AND THE PLATE-TECTONIC FRAME

CONTEXT + EXACT CORE

Plate tectonics explains why most strong earthquakes and active volcanoes form narrow belts rather than a random global scatter.

MECHANISM / ARGUMENT

The Himalaya is intensely seismic because of continent–continent shortening, yet it lacks the continuous oceanic-slab arc volcanism typical of the Andes or Japan.

CONSEQUENCE / CONTRAST

The best explanatory chain is internal energy → plate movement → stress or melting → event → landform/hazard.

UPSC TRAP / ANSWER USE

Plate tectonics explains why most strong earthquakes and active volcanoes form narrow belts rather than a random global scatter.

ANSWER LINE: Plate tectonics explains why most strong earthquakes and active volcanoes form narrow belts rather than a random global scatter.

02

STRESS, STRAIN, ELASTIC REBOUND AND FAULTS

CONTEXT + EXACT CORE

Fault maps identify where rupture has occurred or may be geologically possible; they do not by themselves provide a date-specific prediction.

MECHANISM / ARGUMENT

Fault maps identify where rupture has occurred or may be geologically possible; they do not by themselves provide a date-specific prediction.

CONSEQUENCE / CONTRAST

Elastic strain is recoverable until failure, whereas permanent deformation remains.

UPSC TRAP / ANSWER USE

Stress is force per unit area; strain is the deformation produced.

ANSWER LINE: Elastic rebound converts long-term tectonic loading into short-duration ground shaking, explaining how gradual plate motion produces a sudden disaster trigger.

03

EARTHQUAKE TERMINOLOGY, CAUSES AND CLASSIFICATIONS

CONTEXT + EXACT CORE

It does not mean every reservoir causes a damaging earthquake.

MECHANISM / ARGUMENT

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MECHANISM / ARGUMENT

It does not mean every reservoir causes a damaging earthquake.

CONSEQUENCE / CONTRAST

Most damaging earthquakes are shallow because the source is closer to the surface, but damage is not fixed by depth alone.

UPSC TRAP / ANSWER USE

"Human-induced" does not mean humans create tectonic stress from nothing.

ANSWER LINE: Focal depth is a source descriptor, whereas disaster severity is a relational outcome produced by source, path, site, exposure and vulnerability.

04

FOCUS, EPICENTRE, SEISMOGRAPHS AND LOCATING AN EVENT

CONTEXT + EXACT CORE

Modern location uses many stations, a velocity model and iterative computation.

MECHANISM / ARGUMENT

The earliest small motion is normally the P arrival, followed by S and then longer-period surface waves.

CONSEQUENCE / CONTRAST

Three-circle triangulation is the learner model, not the full operational method.

UPSC TRAP / ANSWER USE

Modern location uses many stations, a velocity model and iterative computation.

ANSWER LINE: Epicentral location is a travel-time inference: arrival differences yield distance, multiple stations yield position, and additional modelling estimates depth and rupture geometry.

05

P, S, LOVE AND RAYLEIGH WAVES

CONTEXT + EXACT CORE

S waves require shear rigidity and therefore do not pass through a liquid.

MECHANISM / ARGUMENT

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CONSEQUENCE / CONTRAST

P waves can pass through both solid and liquid but change velocity and direction at material boundaries.

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CONSEQUENCE / CONTRAST

P waves can pass through both solid and liquid but change velocity and direction at material boundaries.

UPSC TRAP / ANSWER USE

"Surface waves are most destructive" is a useful rule, not an absolute law.

ANSWER LINE: Arrival order is P first, S second and surface waves later; medium control is P through solids and liquids but S through solids only.

REFRACTION, DISCONTINUITIES AND SHADOW ZONES

CONTEXT + EXACT CORE

The reappearance of P phases beyond the shadow zone results from strong refraction through the core.

MECHANISM / ARGUMENT

The reappearance of P phases beyond the shadow zone results from strong refraction through the core.

CONSEQUENCE / CONTRAST

Do not turn a classroom shadow-zone sketch into an exact operational travel-time model.

UPSC TRAP / ANSWER USE

This is not evidence that the whole core is liquid; other wave evidence supports a solid inner core within the liquid outer core.

ANSWER LINE: The larger S-wave shadow is a state-of-matter test, while the P-wave shadow is principally a refraction geometry.

MAGNITUDE, INTENSITY AND LOGARITHMIC ENERGY

CONTEXT + EXACT CORE

Intensity is not a synonym for economic loss because two buildings at the same intensity can perform differently.

MECHANISM / ARGUMENT

M_w converts this physical source measure to a logarithmic magnitude.

CONSEQUENCE / CONTRAST

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UPSC TRAP / ANSWER USE

"Richter scale" properly refers to local magnitude developed for a particular instrumental and regional context.

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ANSWER LINE: Magnitude measures the earthquake; intensity measures the earthquake-at-a-place.

08

GLOBAL EARTHQUAKE DISTRIBUTION AND PLATE MARGINS

CONTEXT + EXACT CORE

Ridge earthquakes are shallow because the lithosphere is thin and hot.

MECHANISM / ARGUMENT

Its location reinforces the active western-Pacific plate-boundary belt; it does not prove that every Pacific earthquake produces a tsunami.

CONSEQUENCE / CONTRAST

Bhuj, Latur and New Madrid show that plate interiors are relatively—not absolutely—stable.

UPSC TRAP / ANSWER USE

The Circum-Pacific belt is the dominant global concentration of subduction-related earthquakes and volcanoes.

ANSWER LINE: Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

09

SECONDARY HAZARDS, SITE RESPONSE AND THE RISK EQUATION

CONTEXT + EXACT CORE

Hazard is physical possibility; vulnerability is susceptibility; risk is their interaction with exposure; disaster is realised severe disruption beyond coping capacity.

MECHANISM / ARGUMENT

Urban disaster risk is often a network problem: a structurally surviving hospital may still fail functionally if power, water, access or communications collapse.

CONSEQUENCE / CONTRAST

Liquefaction is not "soil melting." It is a temporary loss of effective stress and strength in susceptible saturated granular material.

UPSC TRAP / ANSWER USE

Soft sediment can amplify selected frequencies as waves slow and grow in amplitude.

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TSUNAMI GENERATION, PROPAGATION AND COASTAL IMPACT

CONTEXT + EXACT CORE

In deep water, a tsunami can pass beneath ships with little notice.

MECHANISM / ARGUMENT

"Move to 10 metres" is not a universal rule; follow official local maps and routes.

CONSEQUENCE / CONTRAST

In deep water, a tsunami can pass beneath ships with little notice.

UPSC TRAP / ANSWER USE

Strong or long coastal shaking, unusual sea-level change or an official alert requires immediate movement along identified evacuation routes.

ANSWER LINE: Strike-slip movement with little vertical displacement is less efficient at generating a large basin-wide tsunami than thrust displacement, although landslides triggered by any event may create a local tsunami.

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VOLCANO ANATOMY, MAGMA, LAVA, SILICA, GAS AND VISCOSITY

CONTEXT + EXACT CORE

A caldera is a much larger collapse structure commonly related to withdrawal of a substantial magma volume.

MECHANISM / ARGUMENT

Dissolved volatiles exsolve into bubbles; if viscous magma prevents escape, gas pressure can fragment magma into tephra and drive explosive eruptions.

CONSEQUENCE / CONTRAST

Basaltic magma is generally hotter, less silica-rich and less viscous than andesitic or rhyolitic magma.

UPSC TRAP / ANSWER USE

Water-magma interaction, conduit sealing, eruption rate and external water can change behaviour.

ANSWER LINE: Silica raises viscosity, viscosity impedes degassing, retained gas raises pressure, and pressure-driven fragmentation makes an eruption explosive.

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INTRUSIVE AND EXTRUSIVE FORMS; VOLCANIC PRODUCTS

12

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CONTEXT + EXACT CORE

Batholiths are very large discordant plutonic bodies later exposed by uplift and erosion.

MECHANISM / ARGUMENT

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CONSEQUENCE / CONTRAST

UPSC Prelims 2024 asked whether pyroclastic debris, ash and dust, nitrogen compounds and sulphur compounds are products of volcanic eruptions.

UPSC TRAP / ANSWER USE

It is a flowing mixture of water and volcanic sediment, often channelled down valleys.

ANSWER LINE: The same eruption has a spatially differentiated hazard field—ballistics near the vent, density currents on flanks, ash downwind and lahars down valleys.

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ERUPTION STYLES, VOLCANO TYPES AND ACTIVITY STATUS

CONTEXT + EXACT CORE

A volcano with no historical eruption may still be capable of renewed activity if geological and geophysical evidence shows a live system.

MECHANISM / ARGUMENT

A volcano with no historical eruption may still be capable of renewed activity if geological and geophysical evidence shows a live system.

CONSEQUENCE / CONTRAST

"Active, dormant and extinct" are descriptive status classes, not a compulsory life cycle.

UPSC TRAP / ANSWER USE

Narcondam is volcanic but should not be substituted for Barren Island in the 2018 PYQ.

ANSWER LINE: Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour—UPSC can mix these axes in close options.

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PLATE-SETTING VOLCANISM, RING OF FIRE AND MID-OCEAN RIDGES

CONTEXT + EXACT CORE

The Ring of Fire includes trenches, island arcs, continental arcs, transform segments, earthquakes from shallow to deep, explosive volcanoes and tsunami-generating megathrusts around much of the Pacific margin.

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CONSEQUENCE / CONTRAST

The Mediterranean–Himalayan belt is not one uniform volcanic arc.

UPSC TRAP / ANSWER USE

Continental rifts may begin with normal faulting and basaltic volcanism; continued extension can develop a narrow sea and eventually an ocean basin.

ANSWER LINE: The Ring of Fire is a linked trench–earthquake–arc system, not merely a ring-shaped list of volcanoes.

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MANTLE PLUMES, HOT SPOTS AND FLOOD BASALTS

CONTEXT + EXACT CORE

Deccan volcanism is associated in standard plume interpretation with the Réunion hot-spot system and the Indian Plate's passage, while the Hawaiian chain illustrates a long-lived oceanic hot spot.

MECHANISM / ARGUMENT

Hot spots can supply a reference frame for plate direction and relative rate independent of ridge magnetic stripes, though plume motion and mantle flow complicate a perfectly fixed frame.

CONSEQUENCE / CONTRAST

Deccan volcanism is associated in standard plume interpretation with the Réunion hot-spot system and the Indian Plate's passage, while the Hawaiian chain illustrates a long-lived oceanic hot spot.

UPSC TRAP / ANSWER USE

The Deccan Traps are not a giant single cone.

ANSWER LINE: The deep-plume hypothesis is influential but not the only explanation of intraplate volcanism; lithospheric extension, edge-driven convection and shallow mantle heterogeneity may also contribute.

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CONSTRUCTIVE, DESTRUCTIVE, CLIMATIC AND ENVIRONMENTAL EFFECTS

CONTEXT + EXACT CORE

The climatic effect depends on what reaches the stratosphere, while the regional effect depends on where ash, lava, pyroclastic currents and lahars travel.

MECHANISM / ARGUMENT

Short-lived cooling after a major explosive eruption is principally associated with stratospheric sulphate aerosols, not with ash remaining aloft indefinitely.

CONSEQUENCE / CONTRAST

Ash can eventually contribute nutrients, but fresh ash may initially smother crops, contaminate water and damage machinery.

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CONSEQUENCE / CONTRAST

Ash can eventually contribute nutrients, but fresh ash may initially smother crops, contaminate water and damage machinery.

UPSC TRAP / ANSWER USE

Volcanic landscapes support tourism, geothermal power and mineral resources, but these benefits may also increase settlement and exposure.

ANSWER LINE: The climatic effect depends on what reaches the stratosphere, while the regional effect depends on where ash, lava, pyroclastic currents and lahars travel.

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INDIA'S TECTONIC FRAMEWORK AND REGIONAL SEISMIC RISK

CONTEXT + EXACT CORE

India's earthquake geography must connect tectonic source, geomorphic site and settlement vulnerability; a zone label alone cannot explain loss.

MECHANISM / ARGUMENT

The 2001 Bhuj earthquake demonstrates severe intraplate risk in Kachchh and the role of non-engineered construction and liquefaction.

CONSEQUENCE / CONTRAST

Event counts show monitoring and active deformation, not a deterministic forecast of a larger event.

UPSC TRAP / ANSWER USE

The Peninsular shield is relatively stable, not aseismic.

ANSWER LINE: India's earthquake geography must connect tectonic source, geomorphic site and settlement vulnerability; a zone label alone cannot explain loss.

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INDIA SEISMIC ZONATION, MICROZONATION AND CLASSIFICATION CAUTION

CONTEXT + EXACT CORE

Because the corresponding adopted BIS standard/map could not be independently verified on the public BIS portal at the package cutoff, this package does not silently replace the code baseline with Zone VI.

MECHANISM / ARGUMENT

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CONSEQUENCE / CONTRAST

NCS defines seismic hazard microzonation as classification into zones of relatively similar exposure to earthquake effects and describes it as a GIS-based, site-specific planning tool.

UPSC TRAP / ANSWER USE

Microzonation must feed building regulation, land use, lifeline siting and retrofit priorities.

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Microzonation must feed building regulation, land use, lifeline siting and retrofit priorities.

ANSWER LINE: The familiar BIS macrozonation uses Zones II, III, IV and V; there is no Zone I in that framework.

INSTITUTIONS, RESILIENT CONSTRUCTION, EARLY WARNING LIMITS AND PREPAREDNESS

CONTEXT + EXACT CORE

[CA FACT] NCS June 2026 monthly report: 445 earthquakes were located globally by NCS during the month, with 230 in India and its neighbourhood; a preliminary report covered the 5 June Chamba M5.0 event. [CA FACT] PIB 22 July 2026: 16 June events in Kangra–Chamba; NCS network stated as 172 stations. [CA FACT] PIB 23 July 2026: structural-safety audits, BIS codes, retrofitting and state/ULB enforcement responsibilities. [CA FACT] USGS reviewed events: Mw 7.8 Philippines on 7 June 2026 and Mw 7.7 north-northwest of Ende, Indonesia, on 14 August 2026 illustrate continuing Ring-of-Fire seismicity. [CA FACT] USGS/HVO 10 March 2026: Kilauea episode 43 provides a current volcanic-products and monitoring illustration.

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CONSEQUENCE / CONTRAST

It is not prediction; warning time is short and near-source blind zones are unavoidable.

UPSC TRAP / ANSWER USE

The implementation gap is often the decisive risk multiplier: a technically sound code that is not adopted, supervised or enforced does not create a safe building.

ANSWER LINE: Earthquake resilience is less a problem of knowing that a city is at risk than of converting that knowledge into enforced design, retrofitted lifelines and rehearsed behaviour.